BFIN WS2026 / GROUP ASSIGNMENT

KAPPA FOUNDATION

COMPLETE PROJECT DEVELOPMENT FRAMEWORK

Liability-First, Risk-Constrained Investment Policy with Factor-Based Validation

An integrated, auditable framework derived from the group's original ideas and refined against the official assignment requirements.

DOCUMENT CONTROL	DETAIL
Status	Working development framework — not an executed investment recommendation
Document version	1.0 | Prepared 16 October 2026
Case decision date	31 December 2022
Information cutoff	No later than 31 December 2022 for original investment decisions
Initial submission	22 October 2026, 16:00 (Ho Chi Minh City time)
Revised mandate	23 October 2026, 16:00 (release of new instructions)
Final submission	29 October 2026, 16:00 (Ho Chi Minh City time)
Primary audience	Three-member investment team; course committee and reviewers

Source hierarchy: the official assignment governs; the group responsibility plan and deadlines checklist coordinate delivery; the group ideation document supplies the investment philosophy. Newly proposed methods and illustrative calculations are explicitly identified.

Document roadmap and decision status

This framework specifies how the group should turn its investment ideas into a defensible, reproducible case recommendation. It is a work plan, not a claim that data have been downloaded, models fitted, or a portfolio approved.

HOW TO READ THE FRAMEWORK

LABEL	MEANING
REQUIRED	Direct requirement of the official BFIN case brief [S1].
RETAINED	Concept originating in the group's ideation document [S4].
REFINED	Group concept retained but modified for compliance, analytical validity or execution.
REJECTED	Concept excluded because it conflicts with the permitted investment universe or case constraints.
PROPOSED	Technical implementation choice that must still be agreed by the group.
OPEN	A decision or evidence item that cannot currently be verified from the supplied documents.

Sections

01. Executive direction and investment thesis
02. Source-of-truth case constraints
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05. Seven analytical workstreams and decision gates
06. Scenario architecture and decision-reversal tests
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Appendices A–C: templates, assumptions register and sources

What is already known — and what is not

CONFIRMED FROM CASE	NOT YET ESTABLISHED
$12m initial assets; $2.4m due 2024; $3.6m due 2027.	Approved risk definition, inflation rate and risk-aversion coefficient.
Permitted assets and no leverage/shorting/derivatives.	Final eligible stock/ETF universe and case-date eligibility record.
Historical estimation ends in 2022; 2023–25 evaluated separately.	Estimated returns, full covariance matrices or factor regressions.
Mandatory deliveries and deadlines.	Recommended holdings, funding and 2030 mission-goal probability.

CONTROL NOTE The group must keep analytical outputs and retrospective performance separate from this development framework. The 22 October decision is the first frozen policy record, not a disposable draft.

01 Executive direction and investment thesis

1.1 Core position

RETAINED + REFINED — “Protect the mission, fund the liabilities, invest for sustainable growth.” The foundation's fixed educational commitments take priority over discretionary return-seeking. The residual capital can then pursue real purchasing-power preservation through a permitted, diversified US-equity allocation.

The proposed research design is a Liability-First, Risk-Constrained Investment Policy with Factor-Based Validation. Financial models inform the action; no Sharpe ratio, optimized weight, factor loading or estimated alpha automatically determines the recommendation.

1.2 Seven questions the committee must be able to answer

1. How are the full USD 12 million allocated at the 2022 decision date?
2. Which contractual asset cash flows meet the 2024 and 2027 mandatory payments, at what initial cost?
3. What nominal wealth is needed in 2030 to preserve the real purchasing power of USD 8 million in 2022?
4. What growth return is required, and how much downside risk accompanies the candidate policy?
5. Which policy is recommended, what credible alternative was rejected, and why?
6. What explains the historical returns and risks: market exposure, factors, or estimated alpha?
7. Which measurable evidence, assumption change or trustees' instruction would reverse the recommendation?

1.3 Decision priority and hierarchy

- Mandatory payments cannot be modified; verify their funding before allocating discretionary growth risk.
- Hard permissions (instrument eligibility, no debt, no short selling, no derivatives) are not optimization variables.
- Risk interpretation and purchasing-power objective must be explicit assumptions, not presented as facts chosen by the trustees.
- Compare implementable policies under consistent expected-return inputs and disclosed constraints.
- Explain, challenge and monitor the recommendation; do not promise the 2030 real-wealth goal.

02 Source-of-truth case constraints

2.1 Economic facts at 31 December 2022

CASE ITEM	OFFICIAL CASE FACT
Opening fund	USD 12.0 million unrestricted cash; no other assets, liabilities or committed donations.
2024 obligation	USD 2.4 million on 31 December 2024 (scholarship programme).
2027 obligation	USD 3.6 million on 31 December 2027 (learning centre).
2030 aspiration	Remainder at 31 December 2030 with at least USD 8.0m of December 2022 purchasing power; goal, not promised return.
Risk concern	Trustees are uncomfortable with losses approaching 15% but have not specified measurement or period.
Allowed instruments	Cash; five fictional case bonds; US common stocks; US-listed ETFs investing primarily in US equities.

CASE ITEM	OFFICIAL CASE FACT
Prohibited	Borrowing, short sales, derivatives; no taxes or bond defaults for case bonds.
Case bond terms	USD 1,000 face, annual 31 December coupons from 2023, non-callable; 2022 accrued interest zero.
Historical design	At least 60 common monthly observations ending December 2022, selected within January 2015–December 2022.
Later evaluation	January 2023–December 2025 only for frozen growth-strategy evaluation, not for original 2022 design.

2.2 Bond quotes supplied by the case

BOND	MATURITY	ANNUAL COUPON	PRICE PER $1,000
Alpha	1 year	0.00%	$958.77
Beta	2 years	0.00%	$922.78
Gamma	3 years	4.00%	$1,002.49
Delta	5 years	3.50%	$986.07
Epsilon	8 years	4.50%	$1,040.71

These are fictional executable case quotes, not Yahoo price series. Do not fabricate historical returns or future resale quotations for the case bonds. Source: [S1], Exhibits 1–2.

2.3 Assessment focus

The official marking structure allocates 10 points to mandate/judgement, 20 to bonds/funding risk, 15 to data/reproducibility, 15 to portfolio construction, 15 to factor interpretation, 15 to evaluation/mandate change and 10 to memo/presentation. This framework maps each requirement to a named deliverable and responsible owner rather than treating analysis volume as success.

03 Original ideas: retain, refine, reject

The following is an explicit change log from the group's seven-page ideation paper [S4]. “Retained” means the principle remains; “refined” means its permitted execution or interpretation changes; “rejected” means the original action must not appear as a candidate case allocation.

IDEA FROM GROUP	DISPOSITION	ADJUSTED USE IN THIS PROJECT
Divide $12m by investment purpose	RETAINED	Use funded-payment sleeve, policy liquidity reserve where justified, and residual growth sleeve.
Secure 2024/2027 cash flows	RETAINED	Cash-flow-match liabilities and explicitly model coupons, maturities and reinvestment.
Diversify across securities	REFINED	Diversify by permitted US equity sectors/styles/ETFs and bond/cash sleeves.

IDEA FROM GROUP	DISPOSITION	ADJUSTED USE IN THIS PROJECT
Trade across countries/currencies	REJECTED	Not an investable route under the permitted universe; discuss only as excluded idea.
Crypto/startups/PE/VC/commodities	REJECTED	Outside permitted instruments; no allocation.
Forex/CFDs/Polymarket/derivatives	REJECTED	Outside scope, including explicit derivatives prohibition.
Use bank leverage to boost returns	REJECTED	Borrowing prohibited; design a fully funded long-only policy.
Weak/base/strong market scenarios	REFINED	Use disclosed quantitative shocks; distinguish scenarios from assigned probabilities.
Target $8m nominal in 2030	CORRECTED	Target $8m in 2022 purchasing power: nominal target depends on inflation.
Assume aggressive growth is mandatory	UNPROVEN	Test feasibility and risk; do not decide this before funding calculations.
Select bonds by matching maturity	REFINED	Inspect every contractual payment and compare cash-flow risk before conclusion.
Compare covariance and single-index	RETAINED	Same universe, return forecast and constraints: only risk estimate changes.
Use Sharpe and mean–variance utility	REFINED	Treat utility and return/risk as policy evidence, not automatic decision rules.
Factors and alpha explain growth	RETAINED	Compare market-only and Lecture 7 multifactor models with uncertainty.
Frozen 2023–2025 backtest	RETAINED	Predeclare strategy, benchmark and rebalancing; fully disclose retrospective limitations.
Find reversal threshold	RETAINED	Solve a meaningful switch point and explain how it changes policy.
Final recommendation as investment policy	RETAINED	Provide implementation, review triggers, liquidity, limits and monitoring.

CONTROL NOTE Two precise corrections matter: the 2030 goal is real purchasing power, not an $8m nominal floor; and the group cannot promise the portfolio grows under every scenario.

04 Target investment policy and worked illustration

4.1 Policy architecture

SLEEVE	ROLE	DESIGN RULE
Funding / liability reserve	Pay mandatory $2.4m (2024) and $3.6m (2027).	Document contractual timing, units, cost, coupons and funding status.
Liquidity / operational cash	Avoid forced sales and administer interim coupons, where policy justifies it.	Do not invent a compulsory minimum; any buffer reduces growth capital.

SLEEVE	ROLE	DESIGN RULE
Growth allocation	Pursue 2030 purchasing-power objective.	Permitted US stocks/US-listed equity ETFs only; risk limits and benchmark explicit.

Funding capital + liquidity capital + growth capital = USD 12,000,000. If cash is included within the liability reserve rather than as a separate sleeve, the accounting must still sum without double-counting.

4.2 Real-wealth and required-return equations

Nominal terminal target: W2030,target = USD 8,000,000 × (1 + inflation assumption)^8.

Simplified illustration (only when no additional growth-sleeve contributions or withdrawals occur): required CAGR = (terminal gap / initial growth capital)^(1/8) − 1. Treat external coupons and interim cash according to the declared funding convention.

4.3 Test case: a maturity-aligned candidate, not a recommendation

One analytically useful preliminary configuration is 2,400 Beta zero-coupon bonds and 3,600 Delta bonds, held to maturity. All figures below follow directly from the case quotes; no optimized allocation or market-data estimate is implied.

COMPONENT	INITIAL COST	CONTRACTUAL FUNDING ROLE
2,400 Beta bonds	$2,214,672	Principal $2.4m payable in 2024; no interim coupons.
3,600 Delta bonds	$3,549,852	Principal $3.6m payable in 2027; annual coupon $126,000 in 2023–27.
Residual for growth	$6,235,476	Uncommitted at time zero before any additional cash buffer.
Total	$12,000,000	Arithmetic checks to the initial fund.

Sensitivity example: the FRED 10-year breakeven reading on 30 December 2022 was 2.30%; that is only an information-at-date reference and is not a verified eight-year inflation forecast. Under a hypothetical constant 2.3% inflation rate, 2030 nominal target ≈ $9.596m. With this illustrative funding allocation, an equity-only residual must compound at ≈5.54% per year if Delta coupons are excluded from terminal wealth. If all five $126k coupons are retained as zero-return terminal cash, the residual growth sleeve needs ≈4.64% per year. Neither is a return prediction. [E4]

> **CONTROL NOTE** In the Delta illustration, principal is reserved for the 2027 obligation. Coupons must not be counted simultaneously as funding spending and 2030 terminal cash. The candidate also does not yet constitute an operational liquidity policy.

4.4 Risk measurement rule proposed for board discussion

Translate the informal “near 15%” concern into a primary one-year, withdrawal-adjusted portfolio loss review threshold, with 2022–2030 maximum drawdown and stress losses reported as additional distinct diagnostics. Because trustees have not agreed on a precise risk constraint, initially label 15% as an escalation/review threshold, not a falsely authorized hard limit.

Risk/return preference: U = E(R) − (A/2) × variance(R), with the return horizon and volatility units consistent. Vary A rather than presenting a single arbitrary value as the trustees' risk attitude.

05 Seven analytical workstreams and decision gates

Workstream 1 — Mandate, assumptions and objective design

OWNER: MEMBER 1 | JOINT APPROVAL BY ALL MEMBERS

Question: what must Kappa fund, what would constitute mission success, and which interpretations of risk are defensible?

- Produce the decision-date mandate summary, nominal liabilities, real-value target and investment-permission checklist.
- Propose a dated inflation proxy/central case and conservative alternatives; distinguish survey/market proxies from a forecast.
- Specify a withdrawal-adjusted return calculation, 15% risk interpretation and at least two risk-aversion levels.
- Use a consistent policy-level preference comparison and explicitly explain which points remain trustee decisions.

Outputs: mandate_assumptions.csv, assumptions_register.xlsx/csv, risk_preference_table.csv, policy_notes.md.

> **CONTROL NOTE** Gate WS1: all material assumptions have owners, definition, units, cutoff-era evidence and sensitivity ranges. No post-2022 observation is used to justify the initial 2022 action.

Workstream 2 — Bond valuation, funding and rate risk

OWNER: MEMBER 1 | CROSS-CHECK: MEMBER 3

Question: what can be contractually funded, at what cost, without relying on fictional future bond-market prices?

- Construct a bond-by-year cash-flow matrix, evaluate quoted prices and YTM, and document settlement/coupon conventions.
- Address the incomplete term structure explicitly. Any zero-coupon extrapolation/interpolation is an assumption, not a price uniquely implied by the case.
- Compare maturity-aligned cash-flow funding with a credible competing bond/cash candidate.
- Estimate Macaulay and modified duration and convexity; test small-change approximation against exact repricing.
- Evaluate at least one parallel rate shift and one yield-curve shape shock at documented and defensible magnitudes.
- Separate interim mark-to-market changes from hold-to-maturity ability to pay the 2024 and 2027 obligations.

Outputs: bond_cashflows.csv, bond_yields.csv, funding_coverage.csv, rate_scenarios.csv, duration_convexity_validation.csv.

> **CONTROL NOTE** Gate WS2: liability-date available cash covers both contractual payments under the proposed funding convention; any residual/coupon cash is transparently accounted for.

Workstream 3 — Equity universe, forecast and portfolio design

OWNER: MEMBER 2 | ELIGIBILITY CHECK: MEMBER 1

Question: how should the permitted growth sleeve be invested, and does a simpler benchmark or competing allocation challenge the preferred strategy?

- Select 8–12 US common stocks across at least four sectors plus two US-listed equity ETFs with meaningfully different mandates.
- Prove instrument existence/eligibility by the 2022 decision date; document limitations from ex-post selection and delistings.
- Predeclare a simple investable comparator and a monthly estimation window of at least 60 common observations ending 2022.
- Document expected-return construction; consider moderate shrinkage to mitigate noisy sample means.
- Specify long-only, sum-to-one and any explicit security, sector, ETF, liquidity and trading-cost limits.
- Calculate a recommended and credible rejected policy; rerun the same universe with sample covariance and single-index risk.

- Export actual frozen weights and strategy construction rules to Member 3 before factor modelling.

Outputs: candidate_universe.csv, asset_monthly_returns.csv, expected_returns.csv, weights_recommended.csv, weights_rejected.csv, risk_model_comparison.csv.

> **CONTROL NOTE** Gate WS3: portfolio differences are attributable to the tested changes, not silent changes to inputs. Selected benchmark is investable and not chosen after the 2023–25 outcome.

Workstream 4 — Market-only and multifactor interpretation

OWNER: MEMBER 3 | STATISTICAL REVIEW: MEMBER 2

Question: are the growth sleeve's historical returns explained by market beta, systematic equity factors or estimated abnormal performance?

Minimum design: fit a market-only model and at least one multifactor model covered by Lecture 7 for (i) the frozen growth strategy and (ii) at least two selected securities. Verify the Lecture 7 factor-model menu before finalizing the specification.

Market-only equation: $r_{i,t} - rf_t = alpha_i + beta_{M,i} \times (r_{M,t} - rf_t) + epsilon_{i,t}$.

Candidate three-factor equation (subject to Lecture 7): $r_{i,t} - rf_t = alpha_i + beta_{M,i} \times MKT{-}RF_t + s_i \times SMB_t + h_i \times HML_t + epsilon_{i,t}$.

- Use the same 2015–22 sample and consistent decimal return units for security excess returns and factor variables.
- Report intercept, slope estimates, standard errors, t-statistics or confidence intervals, adjusted R-squared and sample size.
- Interpret magnitude and sign of market, size and value exposures economically; distinguish fit from reliable future outperformance.
- Check whether estimated alpha changes under different factor specifications; disclose residual problems and multiple-testing limits.
- Compare Eta's historical risk-adjusted criterion with Theta's factor/persistence thesis without confusing an alpha estimate with skill.
- Avoid claiming actual 2023–25 alpha persistence without a predeclared diagnostic and appropriate uncertainty language.

Outputs: factor_coefficients.csv, factor_model_comparison.csv, factor_diagnostics.csv, factor_exposures.png, eta_theta_assessment.md.

> **CONTROL NOTE** Gate WS4: each regression table directly supports, weakens or leaves unchanged an investment-policy claim, and the associated uncertainty is visible.

Workstream 5 — Integrated risk and policy comparison

OWNER: ALL THREE | MEMBER 3 VALIDATES INTEGRATION

Question: which complete USD 12m policy better balances fixed obligations, expected terminal wealth and trustee risk concerns?

- Recombine dollar funding/cash sleeves with exact approved growth weights; reconcile to $12,000,000.
- Keep bond market-value risk and growth-equity market risk separate before presenting total policy stress outcomes.
- Compare utility $U(A)$ over a disclosed risk-aversion range, with the same return horizon and risk units.
- Calculate 2030 goal gaps under plausible inflation and return scenarios, without treating scenarios as guaranteed paths.
- Show one chosen policy and one credible rejected policy, with quantified opportunity cost and downside trade-off.

Outputs: full_allocation.csv, integrated_scenario_table.csv, policy_utility_comparison.csv, decision_summary.md.

> **CONTROL NOTE** Gate WS5: selected policy is implementable and defended even when some empirical evidence contradicts it.

Workstream 6 — Decision challenge, threshold and frozen evaluation

OWNER: MEMBER 2 FOR EVALUATION; ASSUMPTION OWNER FOR THRESHOLD; MEMBER 3 VALIDATES

Question: what evidence would make the original policy less credible or force a policy switch?

- Identify a material parameter (inflation, expected growth return, risk aversion or risk limit) that has a measurable policy-switch threshold.
- Use an explicit search/solution procedure rather than reporting sensitivity values that never change the action.
- Freeze the 2022 portfolio construction, benchmark and any predeclared rebalancing rule before viewing 2023–25 results.
- Evaluate the frozen growth strategy against the frozen investable benchmark for 36 months with returns, volatility, drawdown and costs.
- Report underperformance, disappearing alpha, instability and other inconvenient evidence alongside favourable metrics.
- Keep 2023–25 equity evaluation separate from fictional-bond scenarios and from proof of any 2030 goal achievement.

Outputs: decision_threshold.csv, frozen_policy.json/yaml, growth_evaluation_2023_2025.csv, evaluation_commentary.md.

> **CONTROL NOTE** Gate WS6: policy freeze is auditable; a threshold reverses an important choice; the evaluation is not used to reverse-engineer 2022 weights.

Workstream 7 — Governance, reproducibility and final communication

OWNER: MEMBER 3 FOR QA; ALL OWNERS FOR EVIDENCE; ALL FOR RECOMMENDATION

- Maintain one shared Python environment but separate owner notebooks/modules to avoid conflicting edits.
- Record source URLs, acquisition dates, requested periods, tickers/datasets, parameters, failures, resolution and package versions.
- Rebuild all substantive outputs from saved raw data with no internet and no manual data editing.
- Map every committee table/figure to one named machine-generated output and peer reviewer.
- Preserve the exact 22 October original files and a signed/dated consequential-decision log after the revised mandate.
- Prepare memo/slides, contribution statement, individual critiques and disclosed/checked AI assistance.

Outputs: raw/, source_manifest.csv, requirements.txt, README_rebuild.md, qa_report.json, decision_log.csv, memo_assets_manifest.csv.

> **CONTROL NOTE** Gate WS7: a clean offline rebuild reproduces the material numbers and every group member can explain their owned analysis.

06 Scenario architecture and decision-reversal tests

6.1 Scenario families — jointly assessed

FAMILY	ILLUSTRATIVE SHOCK OR TEST	REQUIRED INTERPRETATION
Rates: parallel	Example +100 bps and −100 bps (magnitude must be defended).	Bond repricing; duration/convexity error; funded-payment status.
Rates: curve shape	Example 2-year +50 bps, 5-year −50 bps (illustrative twist).	Term-structure assumption, sensitivity by maturity, residual cash timing.
Growth market	Example one-year −30%, 0%, +20% equity sleeve returns.	Separate sleeve loss from total-fund loss; no assumed “safe growth”.
Inflation	Example 2.0%, 2.3%, 3.0%, 4.0% paths.	2030 nominal target and gap under consistent cash-flow accounting.

FAMILY	ILLUSTRATIVE SHOCK OR TEST	REQUIRED INTERPRETATION
Expected returns	Lower forecast toward benchmark or conservative value.	Feasibility and switching between policies.
Preferences	Vary mean–variance risk aversion A.	Whether the recommended policy changes and at what value.

Shock sizes above are design candidates, not estimates claimed to have been calibrated to 2022 information. The final memo must justify chosen magnitudes. Do not assign probabilities to scenarios unless a documented model and calibration are supplied.

6.2 Decision-reversal logic

For two policies P and Q, under the same expected-return and annual variance conventions: utility_P = mu_P − (A/2)×sigma_P²; utility_Q = mu_Q − (A/2)×sigma_Q². If sigma_P² differs from sigma_Q², the equality threshold is A* = 2(mu_P − mu_Q)/(sigma_P² − sigma_Q²). A threshold only has useful meaning if the sign, feasible preference domain and economic policy switch are checked.

Alternative: find inflation pi* such that projected 2030 wealth equals 8m × (1+pi*)^8 under an explicitly defined fixed policy. “Goal feasible” must be qualified by whether the test uses point forecasts, stress outcomes or modelled attainment probabilities.

6.3 What cannot be promised

- Guaranteed equity returns or guaranteed attainment of real-value goal in 2030.
- Profitability in all weak/base/strong states or downside losses that can never breach 15%.
- Outperformance inferred solely from positive historical alpha or a high backtested Sharpe ratio.
- A genuinely unseen experiment: the case is reconstructed retrospectively in 2026.

07 Data architecture and reproducibility contract

7.1 Shared architecture

```
BFIN_Kappa/
  config/initial_policy.yaml
  data/raw/yahoo/                 # exact downloaded responses
  data/raw/french/                # exact downloaded factor files
  data/processed/                 # deterministic transformations
  data/source_manifest.csv        # provenance and download settings
  notebooks/01_bond_funding.ipynb # Member 1
  notebooks/02_portfolio_evaluation.ipynb # Member 2
  notebooks/03_factor_analysis.ipynb # Member 3
  src/data_io.py  src/returns.py  src/factor_utils.py
  src/qa_checks.py
  outputs/bond/  outputs/portfolio/  outputs/factor/
  tests/  requirements.txt  README_rebuild.md
  run_pipeline.py
```

7.2 Data contract and timing rules

FIELD	SHARED STANDARD
Asset frequency	Monthly simple total returns, including distributions/corporate actions as appropriate.

FIELD	SHARED STANDARD
Internal return units	Decimal (e.g., 1.5% becomes 0.015); French percentage source converted explicitly.
Month index	Calendar month key YYYY-MM; align factor and equity series on matching months.
Estimation	Proposed Jan 2015–Dec 2022; verify ≥60 common monthly returns for every used series.
Evaluation	Jan 2023–Dec 2025; cannot enter original forecasts or optimization.
Initial price boundary	Earlier month-end price may be necessary to compute the first monthly return (e.g., Dec 2014 for Jan 2015).
Dividend/split adjustment	Choose, log and test Yahoo adjusted-price conventions; avoid mixing return definitions.
Missing observations	Log security/period/gap, reason and treatment. No synthetic fills claimed as downloaded data.
Factor source	Kenneth French website, named file, saved raw response and source vintage noted.
Benchmark	Simple investable US equity ETF, chosen using only case-date eligibility/evidence.
Timestamp cutoff	At initial decision, all selection rationale/assumptions must rely on info available by 2022-12-31.
Publication/data revision	Record that latest downloaded returns and factor series may incorporate later revisions.

7.3 Retrieval and offline rebuild requirements

- Provide functioning Python retrieval code invoking yfinance for equities and the Kenneth French website for factors. Do not replace required retrieval with cached workbook datasets.
- Save original bytes or untransformed responses, together with checksums and retrieval date. Record exact yfinance arguments; API end date is exclusive in the documented interface [E2].
- Do not require the internet at analysis rebuild time: the execution path must load saved raw files deterministically.
- Preserve full-precision calculations; separately apply presentation rounding in output tables.
- Record failed requests or corporate-action anomalies and documented resolutions; do not invent missing observations.
- Re-run in a clean environment using the declared package versions and verify material output tables/figures.

7.4 Handoff schema

PRODUCER → CONSUMER	REQUIRED MACHINE-READABLE HANDOFF	VALIDATION
Member 1 → Member 2	funding_allocation.csv: component, amount_usd, payment_date, residual_growth_usd	Full dollar sum equals $12m; funded obligations separately validated.

PRODUCER → CONSUMER	REQUIRED MACHINE-READABLE HANDOFF	VALIDATION
Member 2 → Member 3	frozen_weights.csv: ticker, weight, case_date; growth_returns.csv: month, decimal_return	Weights sum to 1; dates/units/strategy agree with frozen rule.
Member 3 → All	factor_coefficients.csv: strategy_or_ticker, model, alpha, betas, se, n, r2_adj	Same case-date estimation sample and declared monthly factor units.
All → Memo editor	output_manifest.csv: exhibit_id, path, generator, owner, reviewer	Every material exhibit originates from a named reproducible file.

CONTROL NOTE Member 3 owns the shared infrastructure and factor work, not Member 1/2 calculation responsibility. An integration failure is triaged by Member 3, while the analytical owner validates the corrected result.

08 Team ownership, handoffs and governance

8.1 Responsibility matrix

DECISION / ANALYSIS	LEAD	CROSS-CHECK
Mandate/risk/inflation and bond funding	M1	M3 checks bond outputs; all approve policy assumptions
Eligible equity universe/benchmark	M2	M1 eligibility and economic role check
Portfolio optimization and frozen evaluation	M2	M3 checks data/output reproducibility
Market-only and multifactor estimation	M3	M2 reviews estimates and interpretation
Shared data, schemas, environment, offline rebuild	M3	M1/M2 test their modules in the common environment
Policy comparison and utility assumptions	M1 + M2	M3 checks integrated numerics and thresholds
Revised mandate policy response	M1 + M2	M3 reruns/validates changed outputs
Executive recommendation and committee narrative	All 3	One editor ensures consistent language and numbers

8.2 Handoff approval checklist

- The producing member supplies code, input definitions, unit conventions, raw/processed paths and numerical sanity checks.
- The receiving member confirms accepted schema, common months, no missing inputs, and agreed interpretation of each field.
- Any change to weights, funding amounts, universe, estimation dates or policy rules is versioned and recorded.
- Each analysis owner signs off the economic meaning; cross-reviewer records a constructive critique.

8.3 Version control and decision log

The 22 October submission must be preserved intact. Each subsequent material change logs: timestamp, original decision, new decision, new instruction/evidence, files impacted, quantitative delta, decision owner, reviewer and impact on the final recommendation. A revised mandate does not change the 31 December 2022 case date or allow post-cutoff market information in the revised 2022 decision.

09 Critical path, deadlines and final deliverables

9.1 Proposed execution calendar

DATE (2026)	PRIORITY WORK	REQUIRED ACCEPTANCE
16–17 Oct	Agree assumption owners, risk definition and common environment.	Approved conventions; folder skeleton and callable test pipeline.
18–19 Oct	Acquire/log equity and factor raw data; initial funding candidate and equity universe.	Source manifests complete; at least one raw-to-processed smoke test.
20 Oct	Preliminary portfolio allocation and member-3 regression integration.	Handoff schemas work; source/output consistency check.
21 Oct	Freeze candidate initial investment policy and offline-rebuild rehearsal.	Two-page decision record, code and dated data record ready.
22 Oct 16:00	Initial decision submission.	Exact submitted copy preserved; rules explicitly frozen.
23 Oct 16:00	Receive revised trustees' mandate.	Issue-specific change request; original archived unchanged.
24–26 Oct	Quantify revised impact; scenario, factor and evaluation evidence.	Old/new comparison and any revised recommendation documented.
27–28 Oct	Committee memo/slides and clean rebuild; peer critiques.	All figures map to named outputs; owner/reviewer checks signed.
29 Oct 16:00	Archive final deliverables.	Final submission package complete and reproducible.

9.2 Initial decision record — maximum two pages

- Investable eligible candidate universe, selection rationale and 2022 eligibility evidence.
- Estimation window, return definition and investable growth comparator.
- Full $12m initial dollar allocation, mandatory funding plan and growth weights.
- Material inflation, expected-return, trading-cost and risk assumptions.
- Predeclared evaluation and rebalancing rule with transaction treatment.
- Working Python file and dated data record.

9.3 Final archive checklist

- Committee memo: maximum eight pages inclusive of figures/tables; references may occupy one additional page.
- Committee presentation as PDF and editable source.
- Executed Python notebooks/scripts; environment/package specification and concise offline-rebuild instructions.
- Original raw downloads, processed data, generated outputs and documented provenance records.

- Complete 22 October frozen submission and a consequential decision log.
- Contribution statement with substantive contributions and another member's check of each.
- An individual short critique from every member; material AI-use disclosure and verification checks.

9.4 Proposed eight-page final memo budget

MEMO SECTION	PAGES	LEAD
Executive recommendation	0.75	All
Mandate, assumptions and objective	0.75	M1
Bond funding and rate risk	1.50	M1
Equity allocation and robustness	1.50	M2
Market/factor evidence	1.00	M3
Risk, utilities and reversal threshold	0.75	All
Frozen evaluation and revised mandate	1.00	M2 + All
Implementation, monitoring and conclusion	0.75	All

The memo should be organized around the board's decision, not lecture chronology; technical diagnostics and raw data belong in the reproducible supporting archive.

10 QA gates, scoring evidence and risk register

10.1 Non-negotiable verification gates

GATE	PASS CONDITION	FAILURE RESPONSE
Data integrity	Allowed/eligible instruments; historical months/units consistent; provenance retained.	Fix selection/download/return errors; document all exclusions.
Funding	Contractual 2024/2027 cash available with no double-counting.	Redesign bond/cash plan; recalculate residual growth.
Portfolio consistency	Long-only; weights sum; $12m allocation; same risk-comparison inputs.	Resolve constraints or model interface and recompute.
Factor validity	Relevant models/subjects; correct excess returns; uncertainty shown.	Recheck Lecture 7 and regressions; rewrite overclaims.
Temporal integrity	2023–25 untouched in original policy estimation and selection.	Revert retrospective contamination; document correction.
Offline reproducibility	Fresh clean run reproduces every material exhibit without internet.	Repair raw/schemas/paths/versions; rerun entire pipeline.
Committee readiness	One clear recommendation, one credible rejection, trade-offs, reversal evidence.	Rework decision story rather than adding unsupported complexity.

10.2 Official assessment-to-evidence map

MARKING AREA	MARKS	WHAT WILL BE VISIBLE
Mandate and judgement	10	Defined 15% concern, inflation, policy utility, stated uncertainty.
Bond/funding risk	20	Pricing/YTM, term curve, duration/convexity, exact parallel/shape repricing, payments.
Data/reproducibility	15	Raw records, logged settings, code, date alignment, offline rebuild.
Portfolio/robustness	15	Eligible universe, benchmark, two policies, covariance vs single-index, limits.
Market/factor explanation	15	Growth + two securities; two model classes; alpha uncertainty/Eta–Theta review.
Evaluation/mandate change	15	Frozen 2023–25 comparison, adverse evidence, quantified policy reversal/change.
Committee communication	10	Concise board rationale, traced figures, slide source, critiques and disclosure.

10.3 Key execution risks and mitigations

RISK	WHY IT MATTERS	CONTROL
Unapproved portfolio scope	Crypto/FX/CFD/leverage contradict case permissions.	Enforce investable universe validation in code and memo.
Survivorship or selection hindsight	2026 review may favor 2022 survivors and later winners.	Case-date eligibility evidence and explicit retrospective disclosure.
Factors and Yahoo return mismatch	Different corporate-action and factor computation conventions.	Separate raw sources; document methodology; align dates/units.
Integration dependency on M2	Factor portfolio regression needs actual frozen weights/returns.	Test factor code early; require dated schema handoff.
Unstable optimized portfolio	Sample means/covariances are noisy.	Constraints, shrinkage sensitivity and simple benchmark comparator.
Cash-flow double-counting	Coupons may wrongly fund two objectives.	Single cash ledger and reconciling funding/terminal calculations.
No offline rebuild	Working online notebook does not satisfy submission.	Clean-network-off rehearsal and deterministic pipeline.
Overstated results/AI dependence	Statistical fit mistaken for forecasts; integrity risks.	Owner explanations, peer checks and precise AI disclosure.

10.4 Go / no-go rule for the recommendation

The final recommendation is defensible only if (1) it complies with all instrument restrictions, (2) funds both non-cancellable obligations by documented contractual/cash mechanics, (3) states the uncertain 2030 goal as a goal rather than guarantee, (4) compares a credible alternative under explicit preferences and costs, (5) survives core data/model checks or candidly discloses where it does not, and (6) provides observable reasons to revisit the policy.

Appendix A — Proposed initial assumption register

All entries below are development proposals or open decisions. They are not statements that Kappa trustees have authorized the values.

ASSUMPTION	STARTING POSITION	OWNER / STATUS
Historical estimation	Jan 2015–Dec 2022 (96 monthly observations) if data permit; 60-month sensitivity.	M2 + M3 / PROPOSED
Equity benchmark	Broad US equity ETF, named before 2023–25 outcomes.	M2 / OPEN
Inflation central input	Evaluate cutoff-dated data; 2.3% breakeven is a candidate reference, not 8-year certainty.	M1 / OPEN
15% concern	One-year withdrawal-adjusted total-fund loss review trigger; also report drawdown.	M1 + All / PROPOSED
Cash/liquidity minimum	Evidence-based buffer if useful; none presumed by case.	M1 / OPEN
Stock/security limits	Propose explicit long-only security and sector concentration limits.	M2 / OPEN
Trading costs	Defend a simple 2022-consistent turnover-cost assumption; test alternatives.	M2 / OPEN
Factor specification	Market-only plus Lecture 7 confirmed multifactor regression.	M3 / OPEN
Rebalancing	Frozen buy/hold or rule-based schedule with predeclared trading costs.	M2 + All / OPEN
Policy threshold	One switch threshold with quantified sign, economic meaning and feasibility.	Assumption owner + M3 / OPEN

Appendix B — Templates for project records

B1 Source record — minimum fields

source_name | download_url | retrieval_datetime | ticker_or_dataset | requested_start | requested_end | frequency | yfinance_args_or_factor_archive | package_version | raw_filepath | file_sha256 | missing_data_notes | failures_and_resolution | owner

B2 Decision log — minimum fields

decision_id | decision_timestamp | original_22Oct_policy_reference | prior_choice | new_choice | source_of_change | quantified_delta | information_cutoff_check | affected_outputs | owner | reviewer | rationale

B3 Memo exhibit manifest — minimum fields

exhibit_id | memo_page | table_or_figure_title | output_filename | generating_script | raw_source_ids | data_window | method | owner | reviewer | reproduced_offline

B4 Individual critique prompts

- What is the strongest argument against the final recommended policy?
- Which assumption or dataset limitation materially affects confidence?
- What observable condition would justify reconsidering the allocation?
- How was another member's calculation or interpretation independently challenged?

B5 AI-assistance disclosure requirements

Describe what AI materially assisted (e.g., editing, coding drafts, conceptual checking, visualization), what original sources were independently consulted, which numerical and factual checks humans performed, and who remains accountable for each section. The team must comply with the actual course academic-integrity policy; this framework does not substitute for that policy.

Appendix C — Source and provenance notes

PRIMARY CASE DOCUMENTS — These four sources are the basis for the framework; source notes identify document origin, not independently tested numerical implementation.

- [S1] Kappa Foundation — BFIN WS2026 Group Assignment (official brief), pp. 1–4. Governing case facts, allowed assets, grading, data rules and deadlines.
- [S2] [BFIN] GROUP ASSIGNMENT (1).pdf, pp. 1–7. Group's three-member ownership, handoffs, reproducibility and collaboration plan.
- [S3] [BFIN] GROUP ASSIGNMENT.pdf, pp. 1–2. Group's delivery summary and final-archive checklist.
- [S4] [BFIN] GROUP ASSIGNMENT (2).pdf, pp. 1–7. Group's original investment ideas, risk questions, bond/portfolio/factor reasoning and proposed policy criteria.

EXTERNAL METHODOLOGICAL / SOURCE DOCUMENTATION — Use only 2022-available information to motivate the original 2022 investment decision. Later webpages can explain methods or current software behavior, but they are not permitted as post-2022 forecasting inputs.

- [E1] Kenneth R. French Data Library (factor datasets, descriptions and historical data archive). https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- [E2] yfinance.download reference (documented input arguments, adjusted-price behavior and exclusive end date). https://ranaroussi.github.io/yfinance/reference/api/yfinance.download.html
- [E3] CFA Institute, Fixed-Income Portfolio Management / Liability-Driven Strategies (background methods on cash-flow matching and duration matching). https://www.cfainstitute.org/insights/professional-learning/refresher-readings/2026/liability-driven-index-based-strategies
- [E4] FRED, 10-Year Breakeven Inflation Rate (T10YIE), historical observation on 2022-12-30 = 2.30%; market-based reference not a required inflation assumption. https://fred.stlouisfed.org/series/T10YIE
- [E5] Kenneth French, historical FF3 July archives and published data vintage discussion. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors_archive.html

Final usage note

This is a comprehensive project execution framework, not the final investment-policy result. The team must still select and verify historical securities, retrieve and archive required raw data, perform the Python analyses, confirm the

Lecture 7 factor model, and review/reject candidate allocations based on actual outputs. Retrospective data revisions and ex-post universe selection must remain explicit limitations in the final report.